Bivariate McKay Recurrences and Meromorphic Tensor-Curl Spectral Functions on Spherical Space Forms

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Abstract

Let Γ be a finite subgroup of $\mathrm{Spin}(4) = SU(2)_L \times SU(2)_R$, let $V_n = \mathrm{Sym}^n \mathbb{C}^2$, and let $\boldsymbol{m}_{p,q}$ record the irreducible Γ-multiplicities in $V_p \boxtimes V_q$. If A_L and A_R denote tensor multiplication by the restricted left and right defining representations, we prove

$$\boldsymbol{m}_{p,q} = U_p(A_L/2)U_q(A_R/2)\boldsymbol{e}_0, \qquad [A_L, A_R] = 0.$$

Thus every finite Γ carries a bivariate representation-ring recurrence, without a product assumption. Restriction to the two transverse-traceless helicity diagonals gives distinct rank operators $A_L^2 - 2I$ and $A_R^2 - 2I$ and a diagonal rank–level identity. For a free action on S^3, these vectors are the twisted multiplicities of the normalized rank-J tensor-curl operator.

Classwise multiplicities are quasipolynomial in the level, so each branch spectral function is a finite Hurwitz-zeta combination, meromorphic on $\mathbb{C}$ with possible poles only at $1, 2, 3$. The rank recurrence propagates for all complex s at fixed spectral offset; at the geometric offset $c_J = J+1$, unilateral reindexing contributes an explicit lowest-mode boundary term. The signed branch difference is entire for every finite subgroup. For a free effective action, the remaining poles are central and satisfy $\mathrm{Res}_2 = 0$ and $\mathrm{Res}_1 = -J^2 \mathrm{Res}_3$, yielding a convergent small-time heat expansion and exact relative log-determinant formulas. The cumulative identity $\sum_{r=1}^{J} U_{2r-1} = U_{J-1}U_J$ supplies the corresponding finite polynomial compression used in companion eta-transfer and boundary constructions. The results are confined to round spherical quotients; the nonround problem remains open.

1 Introduction

The spectrum of a round spherical quotient is controlled by two finite pieces of information: the $SU(2)_L \times SU(2)_R$ harmonic content of S^3 and the restriction of those representations to the deck group. The classical spectral literature turns this observation into character formulas and one-variable generating functions [Ike80b, Ike80c, Ike80a, Bär96, LMR21]. The classical McKay correspondence organizes restriction along one $SU(2)$ factor by an affine-ADE adjacency matrix [McK80, Kos85, Ros03, Sut07, Ben16, JWZ20, JWZ21]. For finite subgroups of $SO(4)$, Xiong introduced the corresponding left/right pair as an edge-coloured McKay graph [Xio20].

This paper identifies the spectral recurrence carried by that pair. The basic object is not initially an eta value but the full bivariate multiplicity grid

$$(p,q) \longmapsto \boldsymbol{m}_{p,q} = \left(\dim \mathrm{Hom}_\Gamma(\rho_i, V_p \boxtimes V_q)\right)_{\rho_i \in \mathrm{Irr}(\Gamma)}.$$

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The two Clebsch–Gordan rules act independently on this grid. Their commutativity produces a two-variable Chebyshev reconstruction from a single source at $(0,0)$. The tensor-curl spectrum is then obtained by restricting to the two diagonals $q-p=\pm 2J$. Explicitly,

$$\boldsymbol{m}_{p,q}=U_p(A_L/2)U_q(A_R/2)\boldsymbol{e}_0.$$

This formula places the one-sided affine-ADE recurrences and genuinely two-sided spherical space forms in the same finite algebraic system.

Three distinctions are essential.

(i) The universal theorem is a representation-ring theorem for a finite subgroup of Spin(4). Literal affine-ADE graph language applies to one-sided factors and product reductions, not to an arbitrary two-sided subgroup.

(ii) The positive and negative helicity lines generally propagate with different matrices. Their signed and unsigned assemblies therefore obey a coupled two-matrix identity. The branch laws do not furnish a common single-matrix recurrence, although special invariant-subspace coincidences are not excluded.

(iii) A recurrence of multiplicities does not automatically become the same recurrence for the geometric spectral functions at successive ranks. The geometric eigenvalue is $n+J+1$, so its origin moves with J. The resulting correction is a unilateral boundary term, not a group-cohomological construction.

The main statements are as follows. Theorem 3.1 gives the two-variable representation-ring recurrence, commutation, Chebyshev solution, and rational generating function. Corollary 3.2 records the generic finite cumulative odd-Chebyshev identity, while Remark 3.3 explains its use in the companion eta-transfer and boundary constructions [Mah26a, Mah26b]. Theorem 4.1 restricts it to the tensor-curl helicity lines and proves the diagonal rank–level identity. Proposition 5.3 gives the finite-Hurwitz continuation of every branch. Theorem 7.1 proves all-s propagation at fixed offset, while Theorem 7.4 gives the exact geometric-offset boundary correction. Corollary 8.3 records the relative zeta and log-determinant consequences. Theorem 6.1 proves eta-entirety and identifies all possible pole-producing conjugacy classes, while Proposition 6.2 converts those residues into a convergent heat expansion. Finally, Propositions 8.2 and 8.4 give pole-free relative rows and their exact Hurwitz-derivative form.

Scope. The finite-group representation theory, Clebsch–Gordan identity, one-sided McKay correspondence, character projection, Hurwitz continuation, and round tensor spectra are antecedent machinery. The contribution here is their combination into a bivariate recurrence and its round spectral consequences, rather than a new regularization method. The cumulative Chebyshev identity is purely algebraic. The Dirac-to-tensor-curl eta transfer and its combination with the auxiliary eta, kernel, and Atiyah–Patodi–Singer boundary formulas are established in the companion papers cited above. Extending the present spectral propagation to nonround metrics remains open.

2 Representation and spectral conventions

2.1 The bivariate restriction grid

Let

$$\Gamma<SU(2)_L\times SU(2)_R$$

be finite. Write

$$\mathrm{Irr}(\Gamma) = \{\rho_0, \rho_1, \dots, \rho_r\}, \qquad \rho_0 = \mathbf{1}, \qquad d_i = \dim \rho_i,$$

and let $\boldsymbol{e}_0$ be the coordinate vector of the trivial representation. For $n \geq 0$, put $V_n = \mathrm{Sym}^n \mathbb{C}^2$; thus $\dim V_n = n+1$ and V_n is self-dual. Set

$$W_L = (V_1 \boxtimes V_0)|_\Gamma, \qquad W_R = (V_0 \boxtimes V_1)|_\Gamma.$$

Our McKay-matrix convention is

$$(A_L)_{ij} = \dim \mathrm{Hom}_\Gamma(\rho_i, W_L \otimes \rho_j), \qquad (A_R)_{ij} = \dim \mathrm{Hom}_\Gamma(\rho_i, W_R \otimes \rho_j). \tag{1}$$

Because W_L and W_R are self-dual, both matrices are symmetric nonnegative integer matrices. Define

$$m^i_{p,q} = \dim \mathrm{Hom}_\Gamma(\rho_i, (V_p \boxtimes V_q)|_\Gamma), \qquad \boldsymbol{m}_{p,q} = (m^i_{p,q})_{i=0}^r, \tag{2}$$

and use the zero-extension convention $\boldsymbol{m}_{p,q} = 0$ if $p < 0$ or $q < 0$.

2.2 The round tensor-curl branches

For the unit round S^3, the normalized projected transverse-traceless tensor-curl operator of integer rank $J \geq 1$ has the two $SU(2)_L \times SU(2)_R$ branches

$$\mathcal{H}^+_{J,n} = V_{n+2J} \boxtimes V_n, \qquad \mathcal{H}^-_{J,n} = V_n \boxtimes V_{n+2J}, \qquad n \geq 0, \tag{3}$$

with eigenvalues $+(n+J+1)$ and $-(n+J+1)$ in the orientation used here. Reversing the orientation exchanges the branches and reverses the signed eta function. The local self-gradient construction and its TT compression are standard higher-spin data; we use only the representation-theoretic spectral realization [Hom01, HT21].

We write A_J for the unit-radius self-adjoint realization determined by this orthogonal spectral decomposition on the L^2 space of TT rank-J tensors. Its domain consists of the expansions for which the sum of eigenvalue squared times coefficient norm squared is finite. The displayed spectrum makes its resolvent compact. On a free quotient and in a twist sector, A_J means the corresponding reducing isotypic restriction. In the absolute zeta function below, the negative branch contributes the magnitudes $n + J + 1$; no spectral-cut phase is included.

If the induced action

$$(g_L, g_R) \cdot x = g_L x g_R^{-1}, \qquad x \in SU(2) \cong S^3,$$

is effective and free, the quotient is a round spherical space form. For each ρ_i, the copy-multiplicity convention for the associated flat bundle gives

$$\boldsymbol{b}^+_J(n) = \boldsymbol{m}_{n+2J,n}, \qquad \boldsymbol{b}^-_J(n) = \boldsymbol{m}_{n,n+2J}. \tag{4}$$

The algebraic results below hold for every finite Γ. Freeness is needed only for this manifold interpretation. An element $(g_L, g_R) \notin \{(1,1), (-1,-1)\}$ has a fixed point precisely when g_L and g_R are conjugate in $SU(2)$, equivalently when their defining traces agree.

Remark 2.1 (Central kernel and descent)**.** If $(-1,-1) \in \Gamma$, the action on S^3 is ineffective. A central element $z = (\varepsilon_L, \varepsilon_R)$ acts on $V_p \boxtimes V_q$ by $\varepsilon_L^p \varepsilon_R^q$. Schur's lemma therefore forces $m^i_{p,q} = 0$ unless

$$\rho_i(z) = \varepsilon_L^p \varepsilon_R^q I_{\rho_i}.$$

Every TT line has even total parity $p+q = 2n+2J$ and hence descends as an ordinary representation of the effective quotient by $\langle(-1,-1)\rangle$. Odd-parity spinorial lines instead retain the chosen spin-lift data. No spinorial transfer is used below.

3 The bivariate representation-ring theorem

Theorem 3.1 (Bivariate McKay recurrence)**.** *For all $p, q \geq 0$,*

$$\boldsymbol{m}_{p+1,q} + \boldsymbol{m}_{p-1,q} = A_L \boldsymbol{m}_{p,q}, \tag{5}$$

$$\boldsymbol{m}_{p,q+1} + \boldsymbol{m}_{p,q-1} = A_R \boldsymbol{m}_{p,q}. \tag{6}$$

Moreover,

$$[A_L, A_R] = 0 \tag{7}$$

and the unique quadrant solution with source $\boldsymbol{m}_{0,0} = \boldsymbol{e}_0$ is

$$\boxed{\boldsymbol{m}_{p,q} = U_p(A_L/2) U_q(A_R/2) \boldsymbol{e}_0,} \tag{8}$$

where U_n is the Chebyshev polynomial of the second kind. Equivalently,

$$\sum_{p,q \geq 0} \boldsymbol{m}_{p,q} t^p u^q = (I - tA_L + t^2 I)^{-1} (I - uA_R + u^2 I)^{-1} \boldsymbol{e}_0. \tag{9}$$

Proof. The Clebsch–Gordan identity

$$V_1 \otimes V_p \cong V_{p+1} \oplus V_{p-1}, \qquad V_{-1} = 0,$$

holds also at $p = 0$. Restricting to Γ, tensoring $(V_p \boxtimes V_q)|_\Gamma$ by W_L, and projecting onto ρ_i gives (5) by the definition (1). The right identity is identical with the two factors exchanged.

The matrices are multiplication by $[W_L]$ and $[W_R]$ in the commutative complex representation ring $R(\Gamma)$, proving (7). Since

$$U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x), \qquad U_{-1} = 0, \quad U_0 = 1,$$

the right-hand side of (8) satisfies both recurrences and the quadrant boundary condition. The two three-term recurrences determine the grid uniquely from $\boldsymbol{e}_0$, which proves (8). Summing each recurrence, or using the standard generating series for U_n, gives (9). □

Corollary 3.2 (Cumulative odd-Chebyshev identity)**.** *For every integer $J \geq 1$,*

$$\sum_{r=1}^{J} U_{2r-1}(x) = U_{J-1}(x) U_J(x) \tag{10}$$

as an identity of polynomials in x. Consequently, the left and right representation-ring generators satisfy, separately,

$$\sum_{r=1}^{J} U_{2r-1}(A_L/2) = U_{J-1}(A_L/2) U_J(A_L/2), \tag{11}$$

$$\sum_{r=1}^{J} U_{2r-1}(A_R/2) = U_{J-1}(A_R/2) U_J(A_R/2). \tag{12}$$

Proof. For $x = \cos\theta$ with $\sin\theta \neq 0$, the formula $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$ gives

$$U_J U_{J+1} - U_{J-1} U_J = \frac{\sin((J+1)\theta)[\sin((J+2)\theta) - \sin(J\theta)]}{\sin^2\theta} = U_{2J+1}.$$

Both sides are polynomials, so this identity holds for every x. The case $J = 1$ of (10) is $U_1 = U_0U_1$. If it holds at J, the displayed product identity gives

$$\sum_{r=1}^{J+1} U_{2r-1} = U_{J-1}U_J + U_{2J+1} = U_JU_{J+1},$$

which proves (10) by induction. Evaluating this polynomial identity at $A_L/2$ and at $A_R/2$ proves the two matrix specializations. □

Remark 3.3 (Application to cumulative boundary eta)**.** The transfer theorem of [Mah26a] denotes the symmetric-power multiplication matrices by A_m^X for $X = L, R$. Under the compatible representation-ring conventions, its Clebsch–Gordan recurrence gives $A_m^X = U_m(A_X/2)$, so Corollary 3.2 simplifies the finite odd-rank sums in the cumulative assembly of [Mah26b]. Combining the rankwise transfer with that paper's auxiliary eta, kernel, and Atiyah–Patodi–Singer assembly yields the cumulative boundary eta and reduced correction in the latter paper. The present corollary supplies the polynomial identity; the transfer and boundary theorems retain their respective geometric hypotheses.

Remark 3.4 (Relation to coloured McKay graphs)**.** The pair (A_L, A_R) itself is not new: in coloured-graph form it is the finite-$SO(4)$ construction studied by Xiong [Xio20]. Theorem 3.1 adds the copy-multiplicity grid, the two commuting recurrences, their Chebyshev reconstruction, and the spectral restriction below. For arbitrary Γ, the safest name is a two-sided representation-ring McKay recurrence. An arbitrary two-coloured graph here need not itself be a single affine-ADE diagram.

4 Tensor-curl helicity lines

Put

$$M_+ = A_L^2 - 2I, \qquad M_- = A_R^2 - 2I. \tag{13}$$

For algebraic boundary formulas, we also allow the formal seed $J = 0$ in (4); the geometric tensor-curl family begins at $J = 1$.

Theorem 4.1 (Rank and diagonal rank–level laws)**.** *For every $J \geq 1$ and $n \geq 0$,*

$$\boldsymbol{b}_{J+1}^+(n) + \boldsymbol{b}_{J-1}^+(n) = M_+\boldsymbol{b}_J^+(n), \tag{14}$$

$$\boldsymbol{b}_{J+1}^-(n) + \boldsymbol{b}_{J-1}^-(n) = M_-\boldsymbol{b}_J^-(n). \tag{15}$$

With $\boldsymbol{b}_J^\pm(-1) = 0$, the positive branch also satisfies

$$\begin{aligned} A_LA_R\boldsymbol{b}_J^+(n) = {} & \boldsymbol{b}_J^+(n+1) + \boldsymbol{b}_J^+(n-1) \\ & + \boldsymbol{b}_{J-1}^+(n+1) + \boldsymbol{b}_{J+1}^+(n-1), \end{aligned} \tag{16}$$

and the negative branch satisfies the mirror identity obtained by interchanging L and R.

Proof. Apply (5) twice at $(p, q) = (n + 2J, n)$:

$$A_L^2\boldsymbol{m}_{n+2J,n} = \boldsymbol{m}_{n+2J+2,n} + 2\boldsymbol{m}_{n+2J,n} + \boldsymbol{m}_{n+2J-2,n}.$$

This is (14); the right-factor calculation gives (15).

Applying A_LA_R to $\boldsymbol{m}_{n+2J,n}$ produces the four corners

$$\boldsymbol{m}_{n+2J\pm1,n\pm1}.$$

Reading those corners on the diagonal lines yields (16). Commutation makes the order of the two steps immaterial, and the mirror calculation proves the negative identity. □

Remark 4.2 (Coupled total identities)**.** Put $\boldsymbol{u}_J = \boldsymbol{b}_J^+ + \boldsymbol{b}_J^-$ and $\boldsymbol{v}_J = \boldsymbol{b}_J^+ - \boldsymbol{b}_J^-$. The branch laws give the exact coupled system

$$\begin{aligned}\boldsymbol{u}_{J+1} + \boldsymbol{u}_{J-1} &= \tfrac{1}{2}(M_+ + M_-)\boldsymbol{u}_J + \tfrac{1}{2}(M_+ - M_-)\boldsymbol{v}_J,\\ \boldsymbol{v}_{J+1} + \boldsymbol{v}_{J-1} &= \tfrac{1}{2}(M_+ - M_-)\boldsymbol{u}_J + \tfrac{1}{2}(M_+ + M_-)\boldsymbol{v}_J.\end{aligned}$$

Thus the theorem supplies two branch matrices rather than one matrix closing on either total alone. When $M_+ \neq M_-$ and both $\boldsymbol{u}_J$ and $\boldsymbol{v}_J$ have nonzero components, the coupling terms are visible in the displayed system, so the representation-ring identities alone do not yield a universal common-matrix recurrence. A special solution may nevertheless lie in a common invariant subspace. Spectral propagation below is therefore stated branchwise.

5 Quasipolynomials and finite-Hurwitz continuation

Lemma 5.1 (Classwise quasipolynomiality)**.** *For fixed J, sign $\varepsilon \in \{+,-\}$, and irrep ρ_i, the sequence $n \mapsto b^{\varepsilon}_{J,i}(n)$ is a quasipolynomial of degree at most two. A valid period is the least common multiple of 2 and the orders of the noncentral eigenphases occurring in the two defining slots of the conjugacy classes of Γ.*

Proof. Character projection gives

$$m^i_{p,q} = \frac{1}{|\Gamma|} \sum_{g=(g_L,g_R)\in\Gamma} \overline{\chi_i(g)}\, \chi_p(g_L)\chi_q(g_R). \tag{17}$$

If g_L has eigenvalues $\mathrm{e}^{\pm i\theta}$ with $\theta \notin \{0,\pi\}$, then

$$\chi_p(g_L) = \frac{\sin((p+1)\theta)}{\sin\theta}$$

is periodic in p because g_L has finite order. At $g_L = \pm I$,

$$\chi_p(\pm I) = (\pm 1)^p (p+1),$$

a period-two sign times a linear polynomial. The same alternatives hold in the right slot. On $p = n$, $q = n + 2J$, or with the slots exchanged, the product is polynomial of degree at most two on every common residue class. The finite sum (17) preserves the claim. □

Definition 5.2 (Branch spectral function)**.** For $c > 0$ and $\Re s > 3$, define the vector-valued Dirichlet series

$$\boldsymbol{F}^{\pm}_J(s;c) = \sum_{n=0}^{\infty} \boldsymbol{b}^{\pm}_J(n)(n+c)^{-s}. \tag{18}$$

The geometric offset is $c_J = J + 1$. Componentwise, the signed eta and unsigned absolute spectral zeta functions are

$$\eta_{J,i}(s) = F^+_{J,i}(s;c_J) - F^-_{J,i}(s;c_J), \tag{19}$$

$$\zeta_{|A_J|,i}(s) = F^+_{J,i}(s;c_J) + F^-_{J,i}(s;c_J). \tag{20}$$

Proposition 5.3 (Finite-Hurwitz formula)**.** *Every component of (18) extends meromorphically to $\mathbb{C}$ as a finite linear combination of Hurwitz zeta functions. Its only possible poles are simple poles at $s = 1, 2, 3$.*

Proof. Let $a(n)$ be one component and choose a quasipolynomial period P. For $0 \le r < P$, write on that residue class

$$a(P\ell + r) = \sum_{h=0}^{2} q_{r,h}(P\ell + r + c)^h.$$

This is merely a change of polynomial basis and may have vanishing top coefficients. Then

$$\sum_{n\ge 0} a(n)(n+c)^{-s} = \sum_{r=0}^{P-1}\sum_{h=0}^{2} q_{r,h} P^{h-s}\zeta_{\mathrm{H}}\left(s-h, \frac{r+c}{P}\right). \tag{21}$$

The Hurwitz zeta function is meromorphic with its only pole at first argument 1, proving the assertion. □

Remark 5.4 (Terminology)**.** Some lens-space literature uses "bivariate representations" for one-group highest weights of the form $k\varepsilon_1 + \ell\varepsilon_2$ [LRB18, LMR21]. Here *bivariate* instead refers to the two independent symmetric-power indices (p, q) of $SU(2)_L \times SU(2)_R$.

6 Pole localization, eta entirety, and heat coefficients

For a conjugacy-class representative $g = (g_L, g_R)$, choose folded eigenangles $\theta_L, \theta_R \in [0, \pi]$ so that the defining eigenvalues are $\mathrm{e}^{\pm i\theta_L}$ and $\mathrm{e}^{\pm i\theta_R}$. A class is *central–central* when both angles lie in $\{0, \pi\}$. Write the corresponding signs as $\varepsilon_L, \varepsilon_R \in \{\pm 1\}$ and put $\sigma = \varepsilon_L\varepsilon_R$.

Theorem 6.1 (Eta entirety and central pole localization)**.** *Let $\Gamma < SU(2)_L \times SU(2)_R$ be finite, let $J \ge 1$, and let $c > 0$. For every irreducible coordinate i, the signed function*

$$E_{J,i}(s; c) = F^{+}_{J,i}(s; c) - F^{-}_{J,i}(s; c) \tag{22}$$

is entire.

If the induced action on S^3 is free, all poles of either branch come from central–central classes with $\sigma = +1$; for a free effective action the identity is the only such class. At the geometric offset $c = J + 1$, each branch has

$$\operatorname{Res}_{s=3} F^{\pm}_{J,i}(s; J+1) = \frac{1}{|\Gamma|}\sum_{\substack{C \text{ central}-\text{central}\\ \sigma(C)=+1}} |C|\,\overline{\chi_i(C)}, \tag{23}$$

$$\operatorname{Res}_{s=2} F^{\pm}_{J,i}(s; J+1) = 0, \tag{24}$$

$$\operatorname{Res}_{s=1} F^{\pm}_{J,i}(s; J+1) = -J^2 \operatorname{Res}_{s=3} F^{\pm}_{J,i}(s; J+1). \tag{25}$$

If the action is not free, a noncentral resonant class with $\theta_L = \theta_R = \theta$ contributes additionally

$$\frac{|C|\,\overline{\chi_i(C)}}{|\Gamma|}\frac{\cos(2J\theta)}{2\sin^2\theta} \tag{26}$$

to the residue at $s = 1$ of both *branches. Thus such resonances do not obstruct the entirety of* (22).

Proof. Character projection reduces the statement to one conjugacy class at a time. On a central–central class, the two branch factors are identical:

$$\varepsilon_L^{n+2J}\varepsilon_R^{n}(n+2J+1)(n+1) = \varepsilon_L^{n}\varepsilon_R^{n+2J}(n+1)(n+2J+1),$$

because the exponent $2J$ is even. If $\sigma = -1$, this is an alternating polynomial and every residue-class coefficient mean vanishes. If $\sigma = +1$, then at $c = J + 1$ its polynomial part is

$$(n + 1)(n + 2J + 1) = (n + c)^2 - J^2,$$

which gives (23)–(25).

If exactly one slot is central, the remaining sine quotient is a mean-zero periodic sequence; multiplication by the at-most-linear central factor therefore produces no residue. If neither slot is central, the class sequence is bounded, so only a pole at $s = 1$ is possible. Product-to-sum gives, for the plus branch, a difference of cosines with frequencies $\theta_L - \theta_R$ and $\theta_L + \theta_R$. Folded noncentral angles make the second frequency nonresonant. The first has nonzero mean precisely when $\theta_L = \theta_R$, and its mean is the quantity in (26). Swapping the slots gives the same mean on that locus. A noncentral equality of folded angles is exactly the fixed-vector obstruction to freeness. Hence the branch residue maps are equal for every finite group, proving that their difference is entire, while freeness leaves only the stated central terms. □

The theorem is stronger than regularity of eta merely at the origin: the entire meromorphic pole part cancels between the two helicities. The unsigned function retains local-density information in its central residues.

Proposition 6.2 (Convergent heat expansion)**.** *Let $F(s) = \sum_{n\geq 0} a(n)(n + c)^{-s}$ be any component of a branch function, let P be a quasipolynomial period, and write $r_j = \operatorname{Res}_{s=j} F(s)$ for $j = 1, 2, 3$. Then the first-order heat trace*

$$\Theta_F(\tau) = \sum_{n\geq 0} a(n)\mathrm{e}^{-(n+c)\tau}$$

has the convergent Laurent expansion

$$\Theta_F(\tau) = \sum_{j=1}^{3} \Gamma(j) r_j \tau^{-j} + \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} F(-k)\tau^k, \qquad |\tau| < \frac{2\pi}{P}. \tag{27}$$

All coefficients are finite exact combinations of rational numbers and Bernoulli-polynomial values.

Proof. Split the quasipolynomial into its P residue classes. Each class is a degree-at-most-two polynomial times a geometric series in $\mathrm{e}^{-P\tau}$, hence a rational function whose nearest nonzero singularities occur at $P\tau = \pm 2\pi i$. Its Laurent coefficients are obtained from $\zeta_{\mathrm{H}}(-m, x) = -B_{m+1}(x)/(m + 1)$. Summing the finitely many classes gives (27); equivalently, the negative powers are the residues of $\Gamma(s)F(s)\tau^{-s}$ and the nonnegative powers are its residues at the nonpositive integers. □

7 Meromorphic rank propagation

Theorem 7.1 (Fixed-offset all-s identity)**.** *Fix $c > 0$. For every $J \geq 1$, the following identities hold as identities of vector-valued meromorphic functions on $\mathbb{C}$:*

$$\boldsymbol{F}^{+}_{J+1}(s; c) + \boldsymbol{F}^{+}_{J-1}(s; c) = M_{+}\boldsymbol{F}^{+}_{J}(s; c), \tag{28}$$

$$\boldsymbol{F}^{-}_{J+1}(s; c) + \boldsymbol{F}^{-}_{J-1}(s; c) = M_{-}\boldsymbol{F}^{-}_{J}(s; c). \tag{29}$$

Convention 7.2 (Formal rank-zero seed)**.** The $J = 1$ instance uses $\boldsymbol{b}^{+}_{0} = \boldsymbol{b}^{-}_{0} = \boldsymbol{m}_{n,n}$ as an algebraic seed. A relation among three geometric tensor-curl ranks begins at $J \geq 2$.

Proof. For $\Re s > 3$, the series converge absolutely and (28)–(29) follow term by term from (14)–(15). Each side has the meromorphic continuation of Proposition 5.3. Uniqueness of meromorphic continuation proves the identities on $\mathbb{C}$. □

7.1 The moving geometric origin

Let S and L denote the unilateral right and left sequence shifts:

$$(Sa)(0) = 0, \qquad (Sa)(n) = a(n-1)\ (n \geq 1), \qquad (La)(n) = a(n+1). \tag{30}$$

The symbol L is notation for the left shift; it is not a two-sided inverse to S because the sequences begin at $n = 0$.

Lemma 7.3 (Offset reindexing)**.** *For every quasipolynomial sequence a and $c > 1$,*

$$F_a(s; c+1) = F_{Sa}(s; c), \tag{31}$$
$$F_a(s; c-1) = F_{La}(s; c) + a(0)(c-1)^{-s}, \tag{32}$$

first in the half-plane of absolute convergence and hence meromorphically on $\mathbb{C}$.

Proof. Equation (31) follows by replacing n with $n+1$. For (32), separate the $n = 0$ term and replace $n \geq 1$ by $n - 1$. Proposition 5.3 completes the continuation. □

Theorem 7.4 (Geometric-offset boundary correction)**.** *Let $J \geq 1$, let $c_J = J + 1$, and let $M_\pm$ be as in (13). For each branch, define*

$$\boldsymbol{\mathcal{B}}^{\pm}_J(s) = \boldsymbol{F}_{(S-I)\boldsymbol{b}^{\pm}_{J+1}}(s; J+1) + \boldsymbol{F}_{(L-I)\boldsymbol{b}^{\pm}_{J-1}}(s; J+1) + \boldsymbol{b}^{\pm}_{J-1}(0)J^{-s}.$$

Then, as vector-valued meromorphic functions on $\mathbb{C}$,

$$\boxed{\boldsymbol{F}^{\pm}_{J+1}(s; c_{J+1}) + \boldsymbol{F}^{\pm}_{J-1}(s; c_{J-1}) - M_\pm \boldsymbol{F}^{\pm}_J(s; c_J) = \boldsymbol{\mathcal{B}}^{\pm}_J(s).} \tag{33}$$

At $J = 1$, the $J - 1$ term is the formal algebraic seed; all three terms correspond to geometric tensor-curl ranks when $J \geq 2$. In this boundary case the lowest-mode factor is $1^{-s} = 1$, as intended.

Proof. Apply Theorem 7.1 at the common offset $c = J + 1$. Lemma 7.3 turns the $J + 1$ geometric offset into $S\boldsymbol{b}^{\pm}_{J+1}$ and the $J - 1$ geometric offset into $L\boldsymbol{b}^{\pm}_{J-1}$ plus the displayed lowest-mode term. Subtracting the common-offset identity gives (33). □

Remark 7.5 (What the correction means)**.** The right-hand side of (33) is an exact inhomogeneous telescoping correction, or geometric-offset boundary cocycle. It measures the cost of changing rank and spectral origin simultaneously. The term $\boldsymbol{b}^{\pm}_{J-1}(0)J^{-s}$ is the mode encountered at the unilateral boundary. The word "cocycle" is descriptive here; (33) does not assert a group-cohomology class.

Corollary 7.6 (Signed and unsigned assembly)**.** *The full eta and absolute-zeta vectors are obtained from (33) by taking the branch difference and sum, respectively. The displayed result is the coupled two-matrix identity; a reduction to one matrix would require additional invariant-subspace structure not asserted here.*

8 Relative zeta functions and log-determinants

Let $\boldsymbol{d} = (d_0, \ldots, d_r)^T$ be the dimension vector and let $\boldsymbol{e}_0^T \boldsymbol{z} = z_0$. Define

$$R = I - \boldsymbol{d}\,\boldsymbol{e}_0^T, \qquad (R\boldsymbol{z})_i = z_i - d_i z_0. \tag{34}$$

Thus $R^2 = R$ and $R\boldsymbol{d} = 0$. For each column j of either McKay matrix, taking dimensions in $W_{L/R} \otimes \rho_j$ gives $\sum_i d_i (A_{L/R})_{ij} = 2d_j$. The matrices are symmetric, hence $A_L\boldsymbol{d} = A_R\boldsymbol{d} = 2\boldsymbol{d}$, and both rank matrices satisfy $M_\pm \boldsymbol{d} = 2\boldsymbol{d}$.

Proposition 8.1 (Induced relative recurrence)**.** *On* $\operatorname{im} R$*, define*

$$\widehat{M}_{\pm} = RM_{\pm}R|_{\operatorname{im} R}.$$

For fixed $c > 0$*, the relative branch vector*

$$\widehat{\boldsymbol{F}}_J^{\pm}(s;c) = R\boldsymbol{F}_J^{\pm}(s;c)$$

satisfies

$$\widehat{\boldsymbol{F}}_{J+1}^{\pm}(s;c) + \widehat{\boldsymbol{F}}_{J-1}^{\pm}(s;c) = \widehat{M}_{\pm}\widehat{\boldsymbol{F}}_J^{\pm}(s;c) \tag{35}$$

meromorphically on $\mathbb{C}$*.*

Proof. Apply R to Theorem 7.1. Every vector decomposes as $\boldsymbol{z} = R\boldsymbol{z} + \boldsymbol{d}z_0$. Since $RM_{\pm}\boldsymbol{d} = 0$,

$$RM_{\pm}\boldsymbol{z} = RM_{\pm}R\boldsymbol{z},$$

which proves (35). □

Proposition 8.2 (Pole-free relative rows)**.** *Suppose* Γ *acts freely and effectively on* S^3*. Then every component of* $R\boldsymbol{F}_J^{\pm}(s; J+1)$ *is entire. Consequently the relative branch log-determinant requires no pole subtraction at any stage of its continuation.*

Proof. By Theorem 6.1, the only pole-producing class of an effective free action is the identity. Its character-projection weight in coordinate i is d_i times its weight in the trivial coordinate. The projection R subtracts exactly that contribution. All residues therefore vanish, and Proposition 5.3 leaves no other singularities. □

Corollary 8.3 (Relative branch log-determinant)**.** *The functions in* (35) *are regular at* $s = 0$*. Define the componentwise relative branch log-determinant by*

$$\boldsymbol{\mathcal{L}}_J^{\pm}(c) = -\partial_s \widehat{\boldsymbol{F}}_J^{\pm}(s;c)\big|_{s=0}. \tag{36}$$

Then

$$\boldsymbol{\mathcal{L}}_{J+1}^{\pm}(c) + \boldsymbol{\mathcal{L}}_{J-1}^{\pm}(c) = \widehat{M}_{\pm}\boldsymbol{\mathcal{L}}_J^{\pm}(c). \tag{37}$$

At the geometric offsets, differentiating R(33) *at* $s = 0$ *gives the corresponding inhomogeneous relative log-determinant identity.*

Proof. The possible poles in Proposition 5.3 are $1, 2, 3$, so differentiation at zero is legitimate. Differentiate (35); the geometric-offset statement follows in the same way from Theorem 7.4. □

Proposition 8.4 (Exact derivative form)**.** *Let a component of a relative branch functional be written in the finite-Hurwitz form*

$$\widehat{F}(s) = \sum_{r=0}^{P-1}\sum_{h=0}^{2} \widehat{q}_{r,h} P^{h-s}\zeta_{\mathrm{H}}(s-h, x_r) + \sum_w c_w w^{-s}, \qquad x_r = \frac{r+c}{P}, \tag{38}$$

where the finite point terms record any reindexing used to place the Hurwitz arguments in a chosen fundamental interval. Then

$$-\widehat{F}'(0) = \widehat{F}_z(0)\log P - \sum_{r,h} \widehat{q}_{r,h} P^h \zeta'_{\mathrm{H}}(-h, x_r) + \sum_w c_w \log w, \tag{39}$$

where $\widehat{F}_z(0)$ *denotes the Hurwitz-zeta part of* $\widehat{F}(0)$*. Thus every relative determinant row has a finite exact symbolic decomposition into rational coefficients, Hurwitz-zeta derivatives at rational arguments, and logarithms.*

Proof. Differentiate each finite atom of (38) at zero. The derivative of P^{h-s} supplies the $\log P$ term, while $-\partial_s(c_w w^{-s})|_{s=0} = c_w \log w$. □

For a concrete example, take $\Gamma = Q_8 \times \mathbb{Z}_3$, with Q_8 in the left slot and $\mathbb{Z}_3$ in the right slot. Let E denote the two-dimensional irreducible representation of Q_8, and let c_0, c_1, c_2 denote the characters of $\mathbb{Z}_3$. On the plus branch at $J = 1$, the quasipolynomial period is $P = 12$. Two representative relative rows are

twist	$[\log P](-\widehat{F}_i'(0))$	$\eta_{1,i}(0)$	$-\widehat{F}_i'(0)$
$E \boxtimes c_0$	$-3/4$	$-4/9$	$-0.89893954730931921\ldots$
$E \boxtimes c_1$	$-5/12$	$2/9$	$-0.96202650840501164\ldots$

The eta column is the ordinary signed eta invariant, not the relative invariant. The determinant columns are relative. The decimal column is a display of the exact decomposition (39), not its definition.

Remark 8.5 (Log-determinant scope)**.** For each component, the exponential of (36) is the zeta-regularized determinant ratio of that branch to d_i copies of the trivial branch. The additive recurrence (37) is a log-determinant identity. For arbitrary fixed c, it describes auxiliary shifted spectral products; it is the geometric branch product at rank J only when $c = c_J = J + 1$. The moving geometric family is governed instead by the inhomogeneous identity above. Negative-branch products use absolute eigenvalue magnitudes and contain no spectral-cut phase. These formulas concern the round branch operators; no nonround operator determinant is asserted.

9 Affine-ADE reductions and two-sided examples

9.1 One-sided reduction

Suppose $\Gamma = G \times \{1\}$ with $G < SU(2)$. If N is the ordinary McKay adjacency of G, then

$$A_L = N, \qquad A_R = 2I, \qquad \boldsymbol{m}_{p,q} = (q+1)\boldsymbol{m}_{p,0}.$$

Consequently,

$$M_+ = N^2 - 2I, \qquad M_- = 2I.$$

Thus one TT branch obeys the familiar affine-ADE rank recurrence while the other has vanishing second difference. Both are shadows of the same two-sided theorem. This also explains why the one-sided recurrence technology of [Kos85, Ste85, Spr87, Ste08, Sut07, Ben16, JWZ20, JWZ21] is a direct antecedent but not the full bivariate statement.

9.2 Product reduction

For $\Gamma = G_L \times G_R$,

$$A_L = N_L \otimes I, \qquad A_R = I \otimes N_R,$$

and, after arranging the multiplicities as a matrix,

$$\mathsf{M}_{p+1,q} + \mathsf{M}_{p-1,q} = N_L \mathsf{M}_{p,q},$$
$$\mathsf{M}_{p,q+1} + \mathsf{M}_{p,q-1} = \mathsf{M}_{p,q} N_R^T.$$

If G_L and G_R are binary polyhedral, the factors are affine-ADE adjacencies. A general nonproduct subgroup has no such outer-product matrix arrangement; the vector laws of Theorem 3.1 remain the correct formulation.

9.3 Computational verification

We checked the formulas independently in exact character and Hurwitz-atom arithmetic for the mixed products $Q_8 \times \mathbb{Z}_3$, $2T \times \mathbb{Z}_5$, and $Q_8 \times \mathbb{Z}_5$, together with nonproduct cyclic lifts. The computations recovered the two branch recurrences, Chebyshev reconstruction, dimension identity

$$\sum_i d_i m_{p,q}^i = (p+1)(q+1),$$

orientation exchange, conjugate-twist symmetry, and the finite-Hurwitz decompositions. A deliberately nonfree diagonal $\mathbb{Z}_3$ action exhibited the resonant residue (26) while its signed branch difference remained entire. Direct traces agreed with the heat expansion inside its convergence disk, and the relative determinant examples showed exact pole cancellation and the differentiated geometric-offset identity. These computations test conventions and exceptional cases; the proofs do not depend on their numerical range.

10 Relation to prior work

The representation-theoretic ingredients have several close antecedents. McKay's observation and the work of Kostant organize the one-sided symmetric-power restrictions by affine-ADE data [McK80, Kos85, Ste85, Spr87, Ste08]. Poincaré-series, walk, Cartan, and Chebyshev formulations were developed further in [Ros03, Sut07, Ben16, JWZ20, JWZ21]. Xiong's edge-coloured construction is the nearest antecedent to the two-operator pair [Xio20]; it does not state the bivariate copy-multiplicity recurrence, Chebyshev grid, TT-line system, or a spectral application.

For spherical space forms, Ikeda and successors developed character-theoretic spectral multiplicities and generating functions [Ike80b, Ike80c, Ike80a, IY79, Bär96, LMR21]. Twisted eta formulas, complete finite-$SU(2)$ Dirac tables, and Molien encodings are classical [APS75, Don78, Gil84, CM01, Deg01, Deg09]. Determinants of spherical space forms from Dirichlet-series data are also established [BF08]. To the best of our knowledge, these sources do not state the bivariate copy-multiplicity recurrence with its tensor-curl line system, or the branchwise all-s identity with a moving geometric offset. The monographs of du Val and Conway–Smith [dV64, CS03] concern the classification of finite rotation groups rather than this multiplicity machinery.

11 Discussion

The results separate three layers that are easily conflated.

1. The representation ring determines how sphere harmonics decompose across topology and flat twist. The finite matrices A_L, A_R are a finite exact encoding of this multiplicity layer.

2. The tensor-curl dispersion relation selects the two lines $q - p = \pm 2J$ and assigns the rank-dependent eigenvalue $n + J + 1$.

3. Hurwitz continuation converts the quasipolynomial line data into meromorphic spectral functions and relative branch log-determinants.

The pole theorem adds a fourth structural distinction. The unsigned branch sum retains the central polynomial density, hence the local heat singularities, whereas the signed branch difference discards the entire pole part and retains only regular spectral asymmetry. This is not just regularity

at $s = 0$: eta-entirety holds throughout the complex plane, including in nonfree representation-ring examples where additional resonant poles appear in each branch separately. For free effective quotients, the relative projection removes the remaining identity density and turns every branch row into an entire function. Thus local heat data, global spectral asymmetry, and relative determinants are three different finite images of the same bivariate grid.

At a common offset, the second and third layers preserve the representation-ring recurrence exactly. At the geometric offset, the lowest-mode boundary records the fact that the spectral coordinate changes with rank. This is why the same finite algebra can propagate full spectral functions without erasing their geometric rank dependence.

Several extensions are natural. The bivariate generating function can be used to compare inverse-spectral information carried by the left/right operator pair. In particular, one may ask whether a finite rectangle of $\boldsymbol{m}_{p,q}$, or finitely many TT diagonals, determines the embedded representation ring up to orientation exchange; the present reconstruction reduces that question to finite matrix data but does not solve the resulting equivalence problem. The relative log-determinant recurrence gives an exact route to further round-space-form tables and constrains coincidences across rank: any putative equality must propagate through both branch matrices and the explicit boundary term. A separate nonround problem is to replace the round representation decomposition by a mixed-order projected pseudodifferential analysis on Berger backgrounds. None of those extensions is needed for the present theorems, and the last is not implied by them.

12 Conclusion

The restrictions of $V_p \boxtimes V_q$ to a finite subgroup of Spin(4) form a two-dimensional recurrence governed by two commuting representation-ring operators. Its restriction to the round tensor-curl helicity lines yields two branchwise rank laws and a diagonal rank–level identity. Finite-group character theory makes each line quasipolynomial, so the complete branch spectral function is a finite Hurwitz-zeta object. The rank recurrence then holds meromorphically for all s at fixed offset, while the geometric rank-dependent origin contributes an explicit unilateral boundary correction. Relative zeta functions and determinant ratios inherit the same structure at the log-determinant level.

In addition, the signed eta function is entire for every finite two-sided subgroup, while freeness localizes all unsigned poles to the central polynomial density. At the geometric offset this density has the universal form $(n + J + 1)^2 - J^2$, forcing $\mathrm{Res}_2 = 0$ and $\mathrm{Res}_1 = -J^2\,\mathrm{Res}_3$. These residues generate a convergent heat expansion, and their cancellation under the relative projection gives finite exact formulas for pole-free determinant ratios. The recurrence therefore propagates not only multiplicities but also pole data, heat coefficients, eta functions, and log-determinants, with the same lowest-mode correction whenever the geometric origin moves.

The affine-ADE theory is therefore not replaced but embedded: it is the one-sided or product-factor face of a broader two-sided Spin(4) representation-ring system. This viewpoint turns inverse questions about round tensor-curl spectra into finite problems for a commuting operator pair and its distinguished diagonals. Whether a finite portion of that data determines the embedded representation ring, and how the recurrence deforms away from the round metric, remain open.

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